

SECTION 01

01

The category & the science

What enzymes are, and what the evidence shows

01 The category & the science

03 Hypothesis to test

02 Market & competition

04 The opportunity in food

What are digestive enzymes? How are they different from probiotics?

Digestive enzymes break large food molecules (carbohydrates, proteins, and fats) into small units the gut can absorb. **The body produces most of them itself** in the mouth, stomach, pancreas, and small-intestine lining. Some foods supply enzymes naturally such as pineapple (bromelain), papaya (papain), raw honey, fermented foods, and yogurt's live cultures. Supplements (e.g., Lactaid) matter where that supply runs short.

Digestive enzymes vs probiotics

Digestive enzymes catalysts that act on food

WHAT IT IS

A protein catalyst – not alive. It breaks a specific food molecule apart.

WHEN IT WORKS

Immediately, on the meal it's taken with. Effect is same-meal and dose-dependent.

EVIDENCE PROFILE

Mechanism is direct and measurable – you can show the substrate is broken down.

Probiotics live bacteria that shift the gut

WHAT IT IS

Live microorganisms meant to populate or rebalance the gut microbiome.

WHEN IT WORKS

Over days to weeks, if at all. Taken as ongoing maintenance, not meal-time relief.

EVIDENCE PROFILE

Strain- and person-specific; results inconsistent and hard to attribute to a single dose.

Why might we have digestive enzyme insufficiencies?

Aging

Output declines over time

Endogenous enzyme production (enzymes produced within the body) falls with age

Lactase, in particular, drops for most—adults after childhood which is why ~**65%** of people are **lactose intolerant**

Stress & lifestyle

Secretion gets suppressed

Chronic stress, poor sleep, and irregular eating blunt the nervous-system signals that trigger enzyme release at meals

Disease & gut damage

The machinery is impaired

Pancreatitis, celiac, Exocrine Pancreatic Insufficiency, and post-illness gut damage reduce output

Lead to clinical case for prescription enzymes (PERT, prescription enzyme replacement therapy)

Never produced

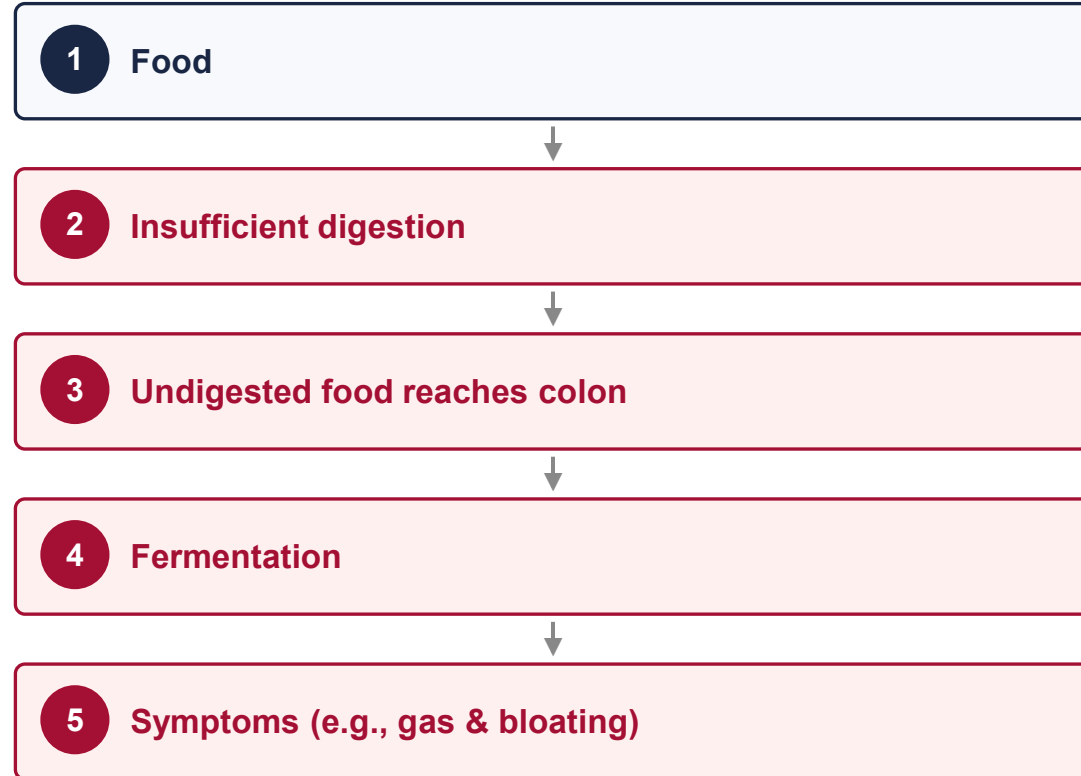
A permanent gap

Some enzymes the body never makes
For these, outside sources are the only option

An example, humans produce no fructan hydrolase at all (found in garlic and onions dishes)

Why do enzyme insufficiencies cause symptoms?

General mechanism



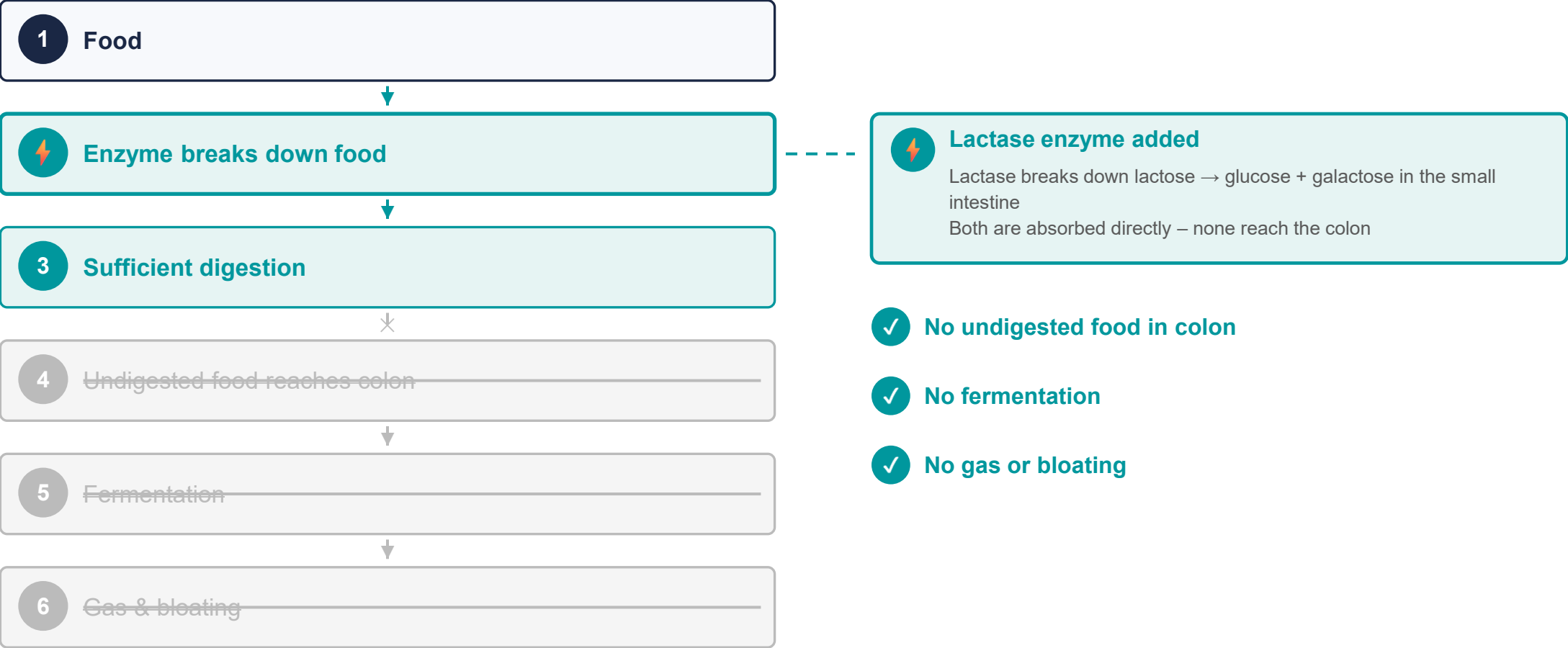
Lactose intolerance – example



Normally, lactase breaks lactose down into smaller, absorbable sugars (glucose + galactose). Without sufficient lactase, **undigested lactose reaches the colon where it is fermented by gut bacteria – producing gas, bloating, cramps and diarrhea.**

How do digestive enzymes work?

With digestive enzyme supplementation



How long have digestive enzymes been around?

Four eras – from treating rare disease to everyday food freedom

1 Medical Era 1900s – 1980s	2 Intolerance Era 1970s – 1990s	3 Wellness Era 2000s – 2010s	4 Precision Era 2020s – PRESENT
Enzyme replacement for disease Pancreatic enzyme replacement (PERT) for cystic fibrosis, pancreatitis & EPI – porcine lipase, protease, amylase. Physician-driven and prescription-only.	Single-food enzyme solutions Replace one missing enzyme so people can eat the food rather than avoid it. The birth of the consumer category.	Broad-spectrum digestive support Multi-enzyme blends for unspecified symptoms, sold through supplement retail. Expanded the market.	FODMAP-specific digestion Match the enzyme to the trigger – fructans, GOS, garlic, onion. Cater to IBS patients. Most react to multiple FODMAP groups.
REPRESENTATIVE Creon · Zenpep · Pancreaze	REPRESENTATIVE Lactaid (1974) · Beano (1990)	REPRESENTATIVE Enzymedica · NOW · Garden of Life	REPRESENTATIVE FODZYME · FODMATE

The industry has moved from treating diagnosed enzyme deficiencies to addressing food sensitivities – each wave **expanding the addressable market** from rare pancreatic disease to **millions of everyday consumers**.

We have all eaten them - why is awareness low?

EVERYDAY FOODS AN ENZYME MADE POSSIBLE

Lactose-free milk & fairlife – lactase is added to the milk; the enzyme has already done its job before you drink it.

Bread & baked goods – amylase and other enzymes improve rise, texture, and shelf life in nearly all commercial bread.

Cheese & beer – rennet sets cheese; enzymes convert starch to sugar in brewing. Ancient, everyday enzyme use.

Lactaid caplets – the one enzyme brand most people can name – and even then, few connect it to a category.

WHY THE CATEGORY IS INVISIBLE

It works silently. A good enzyme produces no sensation – just the absence of a symptom.

It hides in the label. “Lactose-free” sells the outcome, never the enzyme behind it.

No one markets the mechanism. Brands sell relief or food, not the biology – so the concept never enters consumer vocabulary.

What does the evidence say – key findings?

Finding	Detail	Source
01 Enzyme levels decline for many reasons – and some enzymes humans simply don't produce	Aging, stress, disease, and lifestyle all reduce endogenous enzyme output. Humans produce no fructan hydrolase at all – making supplementation the only option for fructan intolerance.	<i>In-vitro study, 2024; Monash FODMAP</i>
02 Enzyme supplementation manages deficiency – but benefit varies by substrate	Carbohydrates show the clearest benefit; protein modest; fat weakest (though evidence exists for all three).	<i>Frontiers ileostomy study, 2024; In-vitro study, 2024</i>
03 Delivery and timing matter – enzymes work best mixed into food at the start of a meal	Direct food contact maximizes enzyme–substrate homogenization at the favorable gastric pH. Pill/capsule formats may isolate the enzyme from its target before it can act.	<i>Monash FODMAP; FODZYME provider brief</i>
04 Most patients react to more than one food trigger – the case for multi-enzyme blends	The average IBS patient is sensitive to ~2.5 FODMAP groups, yet most OTC enzymes target only one. Multi-enzyme blends with fructan hydrolase + lactase + alpha-gal offer broader coverage.	<i>Kaye et al., Gastro Hep Advances 2026</i>
05 Adverse effects are minimal – well tolerated across all studies reviewed	The independently-funded RCT reported zero adverse events over two months. The FODZYME cohort (n=118) also reported no adverse events. Category is FDA-recognized safe for human consumption.	<i>Ullah et al. (2023), Biomed Pharmacother; Kaye et al. 2026</i>

The most defensible evidence anchor: an independently-funded, double-blind, placebo-controlled 2-month RCT showing significant improvement in symptoms, QoL, and sleep with zero adverse events

What does the evidence say – key risks?

Risk	Detail	Source
01 Not every symptom is a food-sensitivity problem	Enzymes only address maldigestion of specific food components – not stress, motility, reflux, or structural causes. If the trigger isn't an undigested substrate, an enzyme won't help.	Functional Dyspepsia RCT
02 Dose is decisive – an underdosed enzyme behaves like placebo	An underdosed enzyme is no better than placebo. Full-dose alpha-galactosidase (300 GALU) improved GOS symptoms; half-dose did not. Substrate load must pass threshold to prevent symptoms.	Monash FODMAP; In-vitro study, 2024
03 Concentrated enzymes can upset sensitive guts	Across competitor reviews, complaints describe new symptoms – a mandate for conservative dosing and clean, FODMAP-free excipients.	Competitor review mining (FODZYME, Enzymedica, Lactaid)
04 Quality and potency vary by source	Manufacturing affects potency and pH/temperature stability; Potential lot-to-lot variability. Standardized sourcing matters.	Functional Dyspepsia RCT

The honest read: enzymes are well-tolerated and FDA-recognized as safe, but they are not a cure-all – they help with food-sensitivity-driven symptoms, and only when correctly matched and dosed. Tolerability and standardized sourcing are formulation requirements.

SECTION 02

02

Market & competition

The category today – size, players, and claims

01 The category & the science

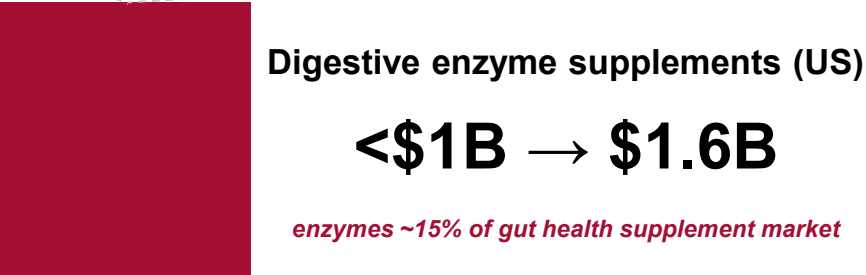
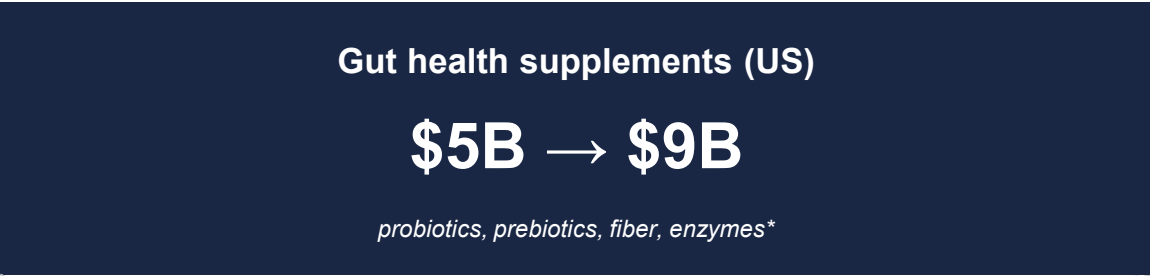
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How big is the market today, and how fast is it growing?

US GUT HEALTH SUPPLEMENT MARKET (BY 2033)



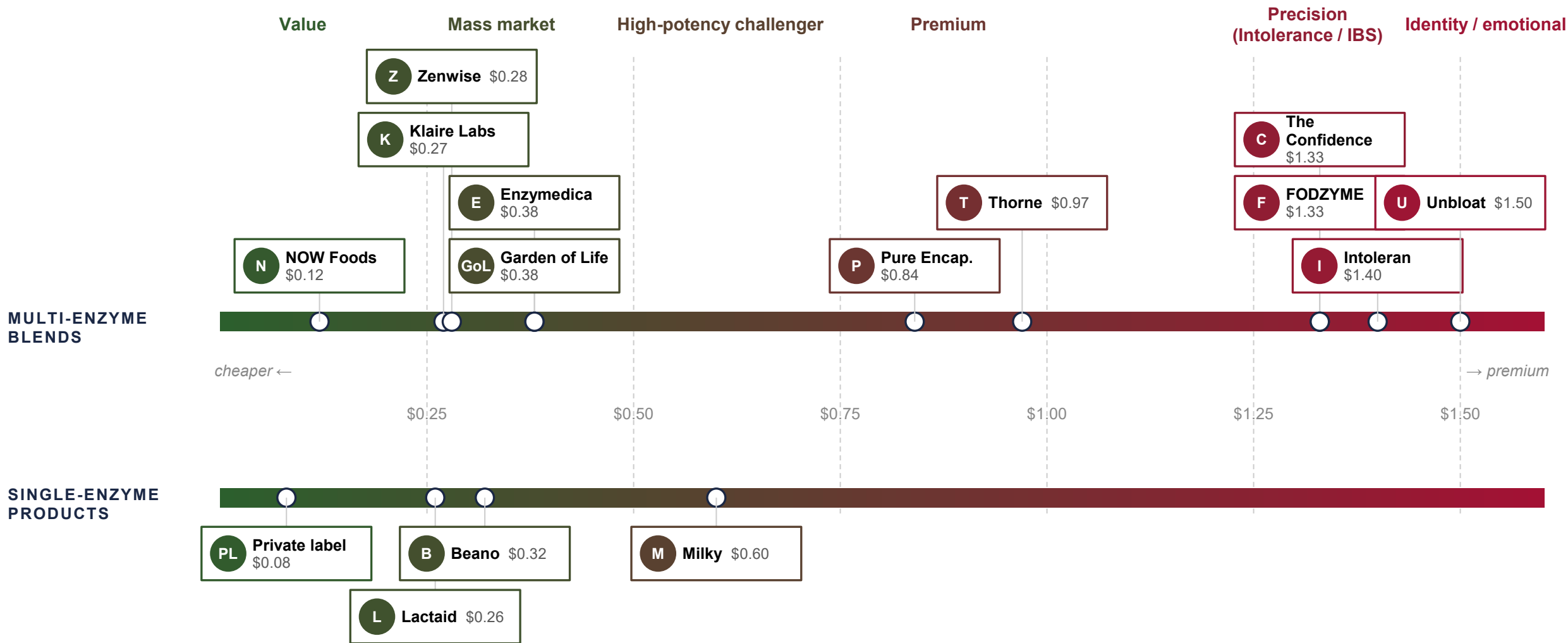
KEY GROWTH DRIVERS

- 1 Rising food sensitivities & avoidance**
Elimination diets are common (e.g., gluten-free, dairy-free).
- 2 Gut health is now an everyday priority**
Microbiome and gut-brain science have entered the mainstream.
- 3 Shift from reactive relief to prevention**
Consumers moving from treating symptoms (e.g., Tums and GasX) to daily routines

Within the supplement market, enzymes are the smallest segment today – but the fastest-growing, expanding faster than the broader digestive-supplement category.

***Not included:** functional food & beverage (the larger gut-health story), prescription enzymes (Creon, ~\$3B, treats EPI), and OTC antacids (Tums/Gas-X, \$3–4B).

Who are the key players – and how do they price per serving?



THE SPREAD Generics and mass blends crowd the floor (~\$0.08–\$0.40); precision and identity brands command a premium (\$1.30–\$1.50)

What are the category archetypes?

GROUP	WHO'S IN IT	PRICE/SERVING	WHERE SOLD	POSITIONING
Value	Private label, NOW Foods	\$0.08–0.12	DTC, Amazon, Retail	Commodity generics. Compete purely on price; no brand, no targeting.
Mass market	Lactaid, Beano, Zenwise, Garden of Life, Enzymedica, Klaire	\$0.26–0.38	DTC, Amazon, Retail	Household names and broad-spectrum blends. Trust and distribution. Accessible.
High-potency challenger	Milky	\$0.60	DTC, Amazon	One enzyme, big dose (lactase ~3.6–7× Lactaid). Discreet format; beats the incumbent head-to-head.
Premium	Pure Encapsulations, Thorne	\$0.84–0.97	DTC, Amazon, Retail	Clinical-grade, clean-label. Quality-seeking buyers.
Precision (Intolerance / IBS)	FODZYME, Intoleran	\$1.33–1.40	DTC, Amazon, Practitioner	Precision FODMAP enzymes; the science leaders (Monash-certified, peer-reviewed). Owns the RD relationship.
Identity / emotional	The Confidence, Unbloat	\$1.33–1.50	DTC, Amazon	Sells confidence and food-freedom to an IBS community. Brand and narrative, not science or IP.

Does anyone have a moat?

FODZYME

Intolerance specialist · \$1.33/serving

POSITIONING

First-mover + proprietary fructan hydrolase + Monash-certified, **peer-reviewed** science.

DIFFERENTIATION

The only commercial fructan hydrolase; a powder you sprinkle on food (format = efficacy); targets fructan, GOS, lactose. Owns the RD relationship.

PATENTS / IP

Patent-**PENDING** (not granted)



The Confidence Co

Identity / emotional · \$1.33/serving

POSITIONING

Brand + food-freedom / anti-stigma narrative + founder (IBS) story + community.

DIFFERENTIATION

16 plant-based enzymes + 3 herbs (broad, not targeted); aimed at women with IBS. Leads with emotion.

PATENTS / IP

None. Defensibility is brand and community only.



Milky

High-potency challenger · \$0.60/serving

POSITIONING

Brand + discreet portable format + high potency + clean-label positioning.

DIFFERENTIATION

Single enzyme (lactase) but a big dose – 10,800–21,000 FCC, ~3.6–7× Lactaid; ultra-thin tablet; FODMAP-free; explicitly beats the incumbent.

PATENTS / IP

None on enzyme, at most design / trade-dress on the sleeve.



Three different bets, no real moat: FODZYME on science (IP only pending), The Confidence Co on identity (no science or IP), Milky on format + potency (a commodity enzyme, bigger dose)

What claims are digestive enzyme companies making?

FOUR CLAIM TYPES

Mechanism Names the specific enzyme & substrate it breaks down. Most defensible, most technical. “breaks down fructans”	Outcome / symptom Promises relief from gas, bloating, discomfort. Concrete but harder to substantiate. “less gas & bloating”	Comparative Positions against an incumbent on potency or speed. Measurable, competitive. “3.6× stronger than Lactaid”	Emotional / identity Sells confidence, freedom, belonging. Broad and vague; no science required. “confidence starts in your gut”
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EXAMPLES

COMPANY	HEADLINE CLAIM	TYPE · HOW IT'S BACKED
FODZYME	“The only enzyme able to target the wide variety of fructans”; “may help break down FODMAPs.”	Mechanism · Monash low-FODMAP certification; peer-reviewed studies
Milky	“Enjoy dairy without discomfort”; “3.6× stronger than Lactaid Original”; fast-acting.	Comparative · Concrete FCC potency numbers; head-to-head vs. incumbent. Narrow (lactose only).
The Confidence	“Confidence comes from within, starting deep inside our gut”; breaks down proteins, carbs, fiber, gluten, fats.	Emotional · Identity-led; broad structure/function. No specific evidence cited.
Lactaid (incumbent)	“Makes dairy easy to digest”; “prevents gas, bloating & diarrhea from lactose.”	Outcome · Decades of category trust; single, well-understood enzyme (lactase).

SECTION 03

03

Hypothesis to test

Is the digestive enzyme opportunity bigger than current usage implies?

01 The category & the science

03 Hypothesis to test

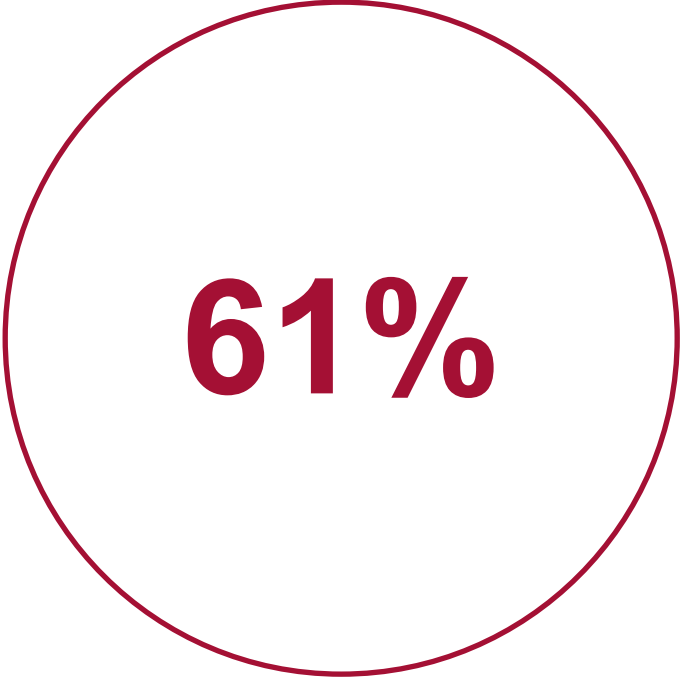
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Is the digestive enzyme opportunity bigger than current usage implies?

61% of people experience **weekly digestive discomfort**
(e.g., gas, bloating, constipation)¹

but only **~10%** have IBS / IBD²



61%

weekly digestive
discomfort



~10%

have IBS / IBD

HYPOTHESIS ON THE DELTA Digestive enzymes may serve more people than use them today – because food sensitivities are common, hard to diagnose, and largely self-managed

1) American Journal of Gastroenterology 113(11):p 1701-1710, November 2018. 2) American Journal of Gastroenterology (<https://gi.org/topics/irritable-bowel-syndrome/>)

How common are food sensitivities? Which enzymes target the sensitivity?

Category	Common foods	Digestive limitation / substrate	Est. US adults affected	Prevalence	Targeting enzyme
FODMAPs	Garlic, onion, wheat	Fructans	~64M	24%	Fructan hydrolase
	Beans, lentils, chickpeas	GOS	Most US adults	Nearly all	Alpha-galactosidase
	Milk, yogurt, ice cream	Lactose	~170M	65%	Lactase
	Apples, mangoes, honey	Excess fructose	~80M	30%	Xylose / glucose isomerase
	Mushrooms, peaches, sugar-free gum	Polyols (sorbitol / mannitol)	~61M	23%	Limited / under development
Other carbs	Sugar, bread, pasta, whole grains	Sucrose, maltose, starch, pectin, insoluble fiber	–	Limited research	Invertase, maltase, amylase, pectinase, cellulase
Protein	Eggs, meat, dairy, legumes	Dietary protein	–	Limited research	Protease
Fat	Oils, butter, fatty meats	Dietary fat	–	Limited research	Lipase

Applied to the U.S. adult population (~267M), these limitations touch tens of millions each – lactose alone implies ~170M. Most people react to more than one group.

Why are food sensitivities underrecognized?

Breath tests have false negatives

Standard hydrogen breath tests carry ~5–15% false negative rates, and results don't reliably predict symptom severity.

Symptoms don't correlate perfectly with tests

Measured malabsorption and reported symptoms are often poorly correlated, making objective thresholds difficult to apply.

Many conditions lack biomarkers

Many have no recognized biomarker; diagnosis relies on symptom-based criteria, leaving a large undiagnosed population.

Self-management is common

Fewer than 20% of people with bloating, abdominal pain, or diarrhea consult a healthcare provider – most manage independently.

SECTION 04

04

The opportunity in food

The larger market

01 The category & the science

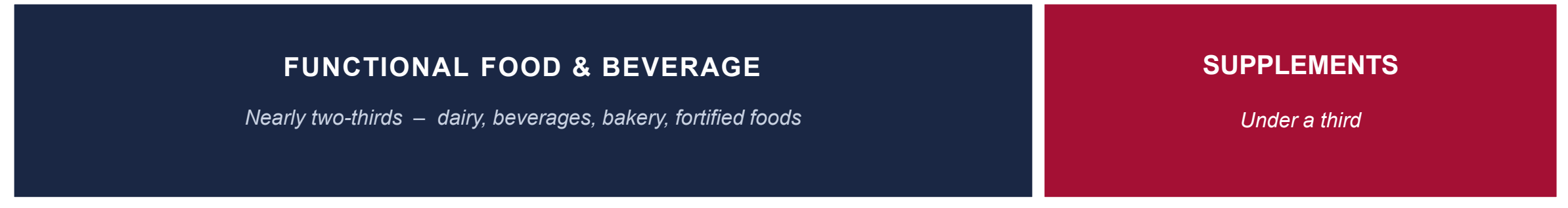
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In gut health, the functional food market is larger than supplements

OVERALL U.S. GUT HEALTH MARKET (FOOD + SUPPLEMENTS) · \$15.8B (2025) → \$30.0B (2033)



In gut health, the market is overwhelmingly functional food

Functional Food: provides health benefits beyond basic nutrition – fortified, enriched, or modified with added probiotics, fiber, enzymes, and other functional ingredients

Pills and powders are a smaller slice of the gut health market

FUNCTIONAL FOOD IS ALREADY WINNING IN GUT HEALTH

Poppi

Prebiotic soda – acquired by PepsiCo for ~\$1.95B (2025). Gut-health benefit, delivered as a drink people crave.



fairlife

Lactase-treated milk – Coca-Cola-owned, \$1B+ in sales.



Sourmilk

Probiotic yogurt – “eat your probiotics.” A supplement benefit delivered as food you enjoy.



Enzyme players fight over the small supplements slice (under a third of the market). The prize is the food side – and it opens into a **\$100B+ functional-food market** in the US by 2032. Delivering enzymes as a food, not a pill, is how to reach it.

Is digestive-enzyme functional food an opportunity worth pursuing?

1

Large, growing market

Enzymes are the fastest-growing supplement segment, sitting beside a functional-food market orders of magnitude larger.

YES

2

Real, painful problem

61% have weekly discomfort; most self-manage. The pain is social, dietary, and quality-of-life – not just physical.

YES

3

Existing solutions fall short

Avoid the food, treat the aftermath, or carry a pill. None lets you simply eat the food you love

YES

4

A defensible way to win

Commodity enzymes have no moat – but food-based delivery has potential for differentiation

THE BET

The problem is painful – and it compounds

THE CASCADE

Symptoms

Gas, bloating, pain, diarrhea after eating



Food avoidance

Cut out trigger foods to prevent symptoms



Reduced quality of life

Narrower diet, social friction, daily anxiety



Symptoms

Gas, bloating, pain, diarrhea after eating

THREE DIMENSIONS OF THE PAIN

● Social

Only ~**45%** are **comfortable discussing** digestive issues; many manage symptoms reactively and in private. Eating out, dating, travel, and shared meals all become sources of stress.

● Quality of life

61% of adults have weekly digestive discomfort. Symptoms drive anxiety around food and a constant vigilance about what to eat.

● Nutrient & diet narrowing

Eliminating whole food groups reduces dietary diversity and can create nutrient gaps – the “fix” (avoidance) creates its own problem.

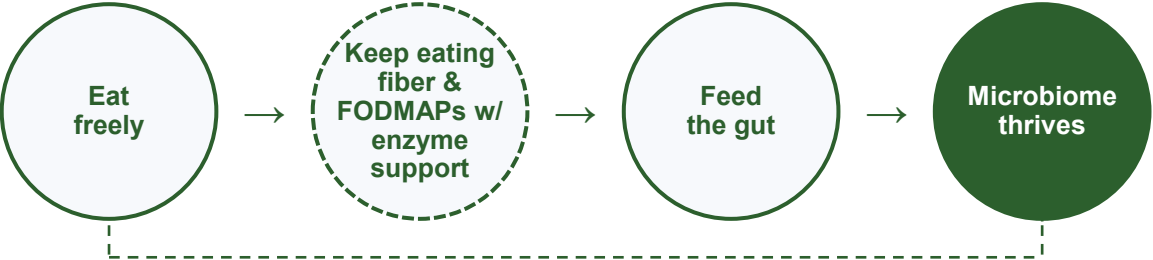
Existing solutions all carry a tradeoff

SOLUTION	HOW IT WORKS	THE TRADEOFF
Elimination / avoidance diets	Cut out trigger foods entirely	Painful and isolating – shrinks the diet, starves gut diversity, hard to sustain, real social cost
Reactive OTC (Tums, Gas-X)	Treat symptoms after they appear	Band-aid – doesn't prevent the problem
Enzyme pills & powders	Take a separate supplement with meals	Remember it, carry it, dose it; “sus” in public; heat-limited
Probiotics	Add bacteria to shift the gut over time	Slow, inconsistent results; strain- and person-specific – often taken as insurance, not relief

Every existing option asks the consumer to change behavior – avoid the food, treat the aftermath, or carry a pill. None lets you simply eat the food you love.

Enzymes have a strong value proposition – and food is the best delivery

THE FOOD FREEDOM LOOP: WHAT ENZYMES UNLOCK



- Enjoyment
- Less fear & anxiety
- Social inclusion
- Nutrients & plant diversity
- Gut microbiome support

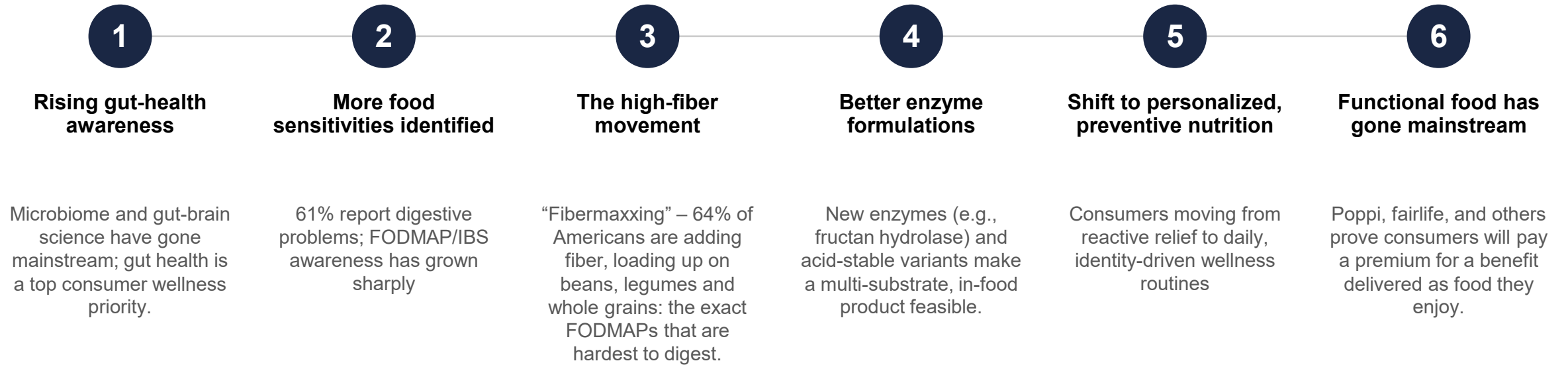
WHY FOOD-BASED ENZYMES?

- 1 More efficacy by design**
Mixing enzymes into food maximizes enzyme–substrate contact – in-food can outperform a pill.
- 2 No behavior to change**
The dose disappears into eating. Nothing to remember or carry – you just eat.
- 3 Socially invisible**
No powder or pill bottle at the table.
- 4 A defensible moat**
A heat- and acid-stable, in-food system is patentable as a process – unlike the commodity enzymes inside it.

Why hasn't everyone adopted enzymes?

Low awareness	Many people don't know enzymes exist or that their symptoms are enzyme-addressable – they assume avoidance is the only option.	→ <i>Food-format + clear mechanism normalizes the category</i>
Format friction	Pills and powders must be remembered, carried, and dosed at the table – easy to forget and awkward in public.	→ <i>Folds the dose into eating – zero friction</i>
Trust & evidence gap	“Supplements don't work” skepticism, vague claims, and a crowded shelf make it hard to know what's real.	→ <i>Mechanism-specific, substantiated claims build trust</i>
Price & value	Premium powders run \$1.30+/serving; commodity pills feel interchangeable – value is unclear.	→ <i>Better per-use value as food, replace expensive elimination diets</i>
Taste & experience	“Tasteless” powders and chalky pills are tolerated, not enjoyed – nothing makes the experience good.	→ <i>Flavor is the point, not a tradeoff</i>

Why now?



Awareness, identification, the fiber surge, formulation science, consumer behavior, and the functional-food category have all shifted

What could go wrong – and how to mitigate it

01 Regulatory & claims risk

As a food, structure/function claims are allowed but **disease claims are not**.

MITIGATION Stay in the safe zone (e.g., “Helps break down lactose / FODMAPs in food”); describe what the enzyme does to the food, never what it does to a disease.

02 Not all symptoms are enzyme-addressable

Enzymes only help food-sensitivity-driven symptoms – not stress, motility, or structural causes.

MITIGATION Target the maldigestion use case clearly; don't over-promise universal relief.

03 Evidence & efficacy varies

Many enzyme claims rest on manufacturer-funded studies; relief varies by person and dose.

MITIGATION Anchor on independent evidence and defensible dosing (e.g., 300 GALU); be clear about scope.

04 Formulation & stability challenge

Enzymes are fragile proteins – heat, acid, and processing can denature them. Food must protect activity through manufacture, shelf life, and the stomach.

MITIGATION Build around heat- and acid-stable enzymes plus encapsulation – the hard problem becomes the process moat.